

Patient based pharma forecasting

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This project estimates patient demand and revenue for a fictional second-line treatment, Therapy A, from 2027 to 2029. It demonstrates Excel modelling, documented assumptions, data review and scenario analysis. All market inputs and treatment assumptions are synthetic. It is a portfolio exercise, not client work or clinical evidence.

Business question

How many patients could start Therapy A, how many remain on treatment, and what revenue follows under different launch assumptions? A commercial analytics reviewer can trace each result to a patient pool, treatment uptake, product share, treatment exits and net price.

Illustrative forecast results

Scenario	2027 revenue	2028 revenue	2029 revenue
Downside	INR 5.10 crore	INR 8.50 crore	INR 8.23 crore
Base	INR 26.57 crore	INR 40.38 crore	INR 34.62 crore
Upside	INR 86.81 crore	INR 124.52 crore	INR 101.71 crore

In the Base scenario, annual revenue rises from INR 26.57 crore to INR 40.38 crore, then falls to INR 34.62 crore. The launch backlog supplies many early starts; continuing patients support year two revenue. As the backlog shrinks, new starts eventually become insufficient to offset exits. This behaviour follows the chosen assumptions and is not a prediction about a real therapy.

What the project demonstrates

Capability	Evidence in the workbook
Data cleaning	Seven preserved source rows; duplicate and missing-population rows excluded with review reasons.
Patient forecasting	Launch pool, incident patients, eligibility funnel, waiting pool and active-patient roll-forward.
Commercial modelling	Uptake, share of new starts, patient-months and net monthly revenue.
Excel controls	One case selector, editable quarterly inputs, formula checks and native revenue chart.

Scope

One fictional indication, one product and five non-overlapping market segments. Twelve calendar quarters are forecast. Revenue excludes taxes, distribution costs and profit; no forecast accuracy claim is made because there are no actual observations.

Patient flow and forecast method

Prevalence supplies a one-time launch pool; incidence supplies new patients each quarter. The model does not add the full prevalent population again every year. The launch pool is assumed to contain patients not already in the modelled second-line treatment pool.

Stage	Calculation and meaning
Market inputs	Included population = 100 million; launch prevalent patients = 105,400; annual incident patients = 20,360.
Incident patients	Annual incidents divided by four, with scenario growth applied at the start of each year. Population remains fixed.
Eligibility funnel	Screening pool × diagnosed fraction × treated fraction × second-line eligibility fraction.
Waiting pool	Opening waiting patients + newly eligible patients – all-therapy starts = closing waiting patients.
Uptake and product share	All-therapy starts = available pool × uptake. Therapy A starts = all-therapy starts × Therapy A share.
Active patients	Opening active patients + Therapy A starts – treatment exits = closing active patients.
Patient months	Continuers × 3 + exiting opening patients × 1.5 + new starters × 1.5, under the default timing assumptions.
Revenue	Patient-months × net monthly price. Annual revenue sums the four matching quarters.

Worked example for the first quarter

Base screening pool = 105,400 + 5,090 = 110,490 patients. Applying 70% diagnosed, 75% treated and 30% second-line eligibility gives 17,402.175 eligible patients. At 18% uptake, 3,132.3915 patients start any therapy. Therapy A captures 25%, or 783.097875 starts.

With no opening Therapy A patients and starts occurring at the quarter midpoint, patient-months = $783.097875 \times 1.5 = 1,174.6468125$. At INR 20,000 per patient-month, first-quarter revenue is INR 23,492,936.25, or approximately INR 2.35 crore. Expected patient counts retain decimals to avoid accumulating rounding errors.

Treatment and epidemiology interpretation

Second-line treatment follows a first-line treatment that fails or stops working. Here, a simple eligibility fraction represents patients reaching that stage; the model does not simulate first-line duration, clinical response, mortality, switches or later treatment lines. The fraction therefore needs clinical validation before any real-world use.

Terminology reference: National Cancer Institute, second-line therapy dictionary entry: <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/second-line-therapy>. The reference supports terminology only, not the synthetic rates.

Assumptions and data review

Driver	Downside	Base	Upside
Diagnosed fraction	60%	70%	80%
Treated fraction	65%	75%	85%
Second-line eligibility	25%	30%	35%
Quarterly uptake	10%	18%	26%
Share of new starts	15%	25%	35%
Quarterly treatment exits	30%	25%	20%
Net monthly price	INR 18,000	INR 20,000	INR 22,000
Annual incidence growth	1%	2%	3%

These are coordinated commercial scenarios, not probability intervals. Every driver has separate inputs for all 12 quarters. Change a later-quarter input to represent a launch ramp or a price change. A single selector determines which assumptions feed the same forecast build.

Review of the synthetic source rows

The original seven rows are retained. The second S03 row duplicates the East segment and is excluded. S06 has no population and is excluded. The five valid segments are assumed mutually exclusive. Include flags document manual review decisions; they must be rechecked when real data is loaded.

Population-weighted SUMPRODUCT formulas calculate prevalent and incident patients. Adding segment percentages directly would give the wrong denominator. Missing population is never estimated from the other regions.

Verification completed

Check	Result and practical meaning
Independent calculation	All three scenarios and three annual revenue totals reproduced independently from the input assumptions.
Formula errors	Workbook scan found no formula errors in the final build.
Reconciliations	Eligible-pool, active-patient and start-allocation differences are zero across all 12 quarters.
Scenario response	Selector changes recalculate patient starts, active patients and annual revenue.
Later-quarter change	Setting the Downside final-quarter price to zero reduced year three revenue; the input was restored.

Important limits

The waiting pool has no mortality or loss-to-follow-up, new starters cannot exit within their starting quarter, and no patient switches or re-entry are modelled. Eligibility happens without a treatment-line delay. No actual sales, patient claims, reimbursement or competitor data calibrates these inputs. These are explicit simplifications, not evidence of clinical forecasting expertise.